

Project Omni-Arb

Intelligent Traingular Arbitrage Framework



THE OPPORTUNITY

In 2026, crypto market efficiency has rendered traditional spatial arbitrage obsolete. Alpha now resides in high-frequency, multi-pair imbalances.

THE MISSION

To engineer an automated system that exploits triangular cycles while using Predictive Analytics to neutralize execution risk.

THE AI-HUMAN TEAM

- **Andrea Cammarano:** Infrastructure & Cloud Scaling
- **Giacomo Lanni:** Quant Strategy & Graph Modeling
- **Edoardo Praderio:** ML Intelligence & Agent Workflow
- **AI Associate (GPT-4o):** Code Optimization & Testing

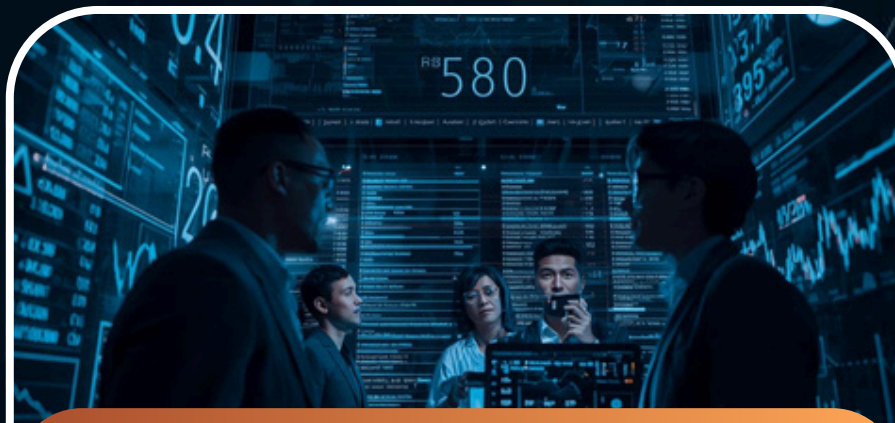
Arbitrage today is not speed-driven, but selection-driven.

The Triple-Check System: From Data to Execution



The Scout (Data Layer)

- WebSocket-based Order Book streaming via CCXT.
- **Real-time monitoring:** Binance, Kraken, Coinbase.
- **Objective:** Lowest latency data ingestion.



The brain (Quant Engine)

- **Logic:** Graph-based Triangular Cycle Detection.
- **Method:** Log-price transformation to identify “Negative Cycles.”
- **Speed:** Detection to execution signal in < 10ms.



The Guardian (ML Advanced Component)

- **Model:** Random Forest Slippage Predictor.
- **Goal:** Filter “Ghost Liquidity” and predict execution success.
- **Confidence:** Trade execution occurs only if $P(\text{success}) > 85\%$.

Process over Product: the Workflow

Section A Agentic Framework

- AI Agents utilized as “Junior Quantitative Developers.”
- Audit Trail: Mandatory [AI-AGENT-CONTRIBUTION] tagging in GitHub.
- Validation: Human-in-the-loop review for all financial logic.

Section B Project Deliverables

- Functional GitHub Repository with full CI/CD traceability.
- Verified “Working Solution” through Paper Trading simulations.
- 8-Page LaTeX Report: Deep-dive into Market Efficiency & Lessons Learned.

Apr 30
Repository & Agentic
Workflow Initialization

May 07
Real-Time Data & Math
Engine Integration

May 14
ML Model Training &
System Backtesting

May 22
Final Documentation &
Repository Freeze

Our competitive edge lies in the intelligence to decide when NOT to trade.